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CSAC 224 - Machine Learning

Customer Segmentation Using K-Means Clustering on Mall Customer Dataset

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1 Introduction

During today's retail and marketing environment, understanding customer behavior was important for business success. Customer segmentation was the process of classifying customer into a distinct groups based on their characteristics or behavior, In this process businesses improve their strategies, enhance the satisfaction of customer, and improved resource allocation. Traditional rule-based segmentation methods often failed to capture the complicated pattern existing in real-word data, making the machine learning approaches increasingly valuable.

This project proposed the implementation of K-Means Clustering, an unsupervised machine learning algorithm, to segment mall customer based on their spending behavior and demographic information. Using the dataset from Kaggle called Mall Customer Segmentation that contained customer age, gender, annual income, and spending score data, the project goal is to identify customer segments that could support data-driven marketing decisions.

One of the issue addressed in this project was the lack of systematic, data-driven approach to customer segmentation in retail environments. Without this approach, business could mismanage marketing budgets, fail to identify high value customers, or can overlook potential market segments. This project used K-Means Clustering to discover hidden patterns in customer data that manual analysis could not reveal.

2 Related work

An important body of research has explored the implementation of clustering algorithms for customer segmentation across different domains. The following related works summarize the key studies that is relevant to this project and identify the gaps that motivated the present study.

CS230: Deep Learning, Winter 2018, Stanford University, CA. (LateX template borrowed from NIPS 2017.)

Aggarwal and Reddy (2014) provided a comprehensive overview of data clustering methods, including K-Means, hierarchical clustering, and density-based methods. Their work built the theoretical foundations that subsequent applied studies were built before, particularly emphasizing K-Means for its scalability and simplicity in handling large datasets. However, their work remained mostly theoretical and did not demonstrate how these clustering methods could be directly applied to retail or marketing settings, leaving a gap in a practical application guidance.

Arora et al. (2016) applied K-Means clustering to retail customer data and demonstrated that segmenting customers by purchasing frequency and monetary value significantly improved targeted marketing campaign effectiveness. Their study emphasizes the importance of feature selection in determining the quality of resulting clusters. However, the study was limited to frequency and monetary features and did not consider demographic variables such as age, gender, or income, which are known to influence spending behavior.

Ezenkwu et al. (2015) inspect the application of K-Means in market segmentation, comparing different initialization strategies including the K-Means++ and random initialization. Their findings showed that K-Means++ consistently gave more stable cluster assignments and lower inertia values. But despite these inputs, the study did not explore how the choice of initialization strategy affects cluster interpretability in a retail business context, unless it validates its result using real-word mall customer data.

Kansal et al. (2018) studied the customer segmentation using the Mall Customer Dataset. They applied K-Means with different numbers of clusters and used the Elbow Method to identify the optimal k=5, identifying distinct customer groups ranging from low-income, low-spending to high-income, high-spending customers. However, their findings relied only on the annual income and spending score characteristics, and did not incorporate age and gender into segmentation, potentially overlooking important demographic patterns in customer behavior.

Syakur et al. (2018) proposed an integration of K-Means with the shape measures for cluster validation, showing that combining clustering with proper evaluation metrics leads to more reliable and interpretable results. Their framework is particularly applicable to this project's methodology. But the study was Performed on a generic dataset and did not demonstrate the approach in the retail or mall shopping context, limiting its practical applicability to customer segmentation problems.

Bhatnagar and Ghose (2004) studied the relationship of customer demographics and online purchasing behavior, finding that age and income are strong predictors of spending behavior. This supports the use of these features in the proposed approach for clustering. However, their study focused on the behavior of e-commerce and did not extend to physical retail settings such as mall environment, where shopping motivation and patterns may differ considerably

Ghosh (2017) studied the gender-based differences in mall shopping behavior, concluding that gender gap analysis can reveal more granular segments, which motivates the inclusion of gender as a feature in this study. However, Ghosh's work deployed descriptive statistical analysis rather than machine learning-based clustering, and therefore did not produce the kind of scalable, data-driven segments that clustering algorithms can make.

Patels and Mehta (2019) applied hierarchical clustering together with K-Means to mall customer data and found that while hierarchical methods provided better visualization cluster through dendrograms, K-Means produced more actionable and well separated segment for business use. Despite this comparison, the study did not analyze clustering quality using quantitative metrics such as the silhouette score, making it hard to objectively identify which method produced more accurate segments.

Liao et al. (2012) inspected the use of dimensionality reduction via Principal Component Analysis before applying K-Means, showing that Principal Component Analysis preprocessing improved cluster separability and lessen computation time on high dimensional datasets. However, the study only focused on high-dimensional datasets and did not assess whether dimensionality reduction provided meaningful benefits for low-dimensional datasets such as the Mall Customer Datasets, which only contains five attributes.

Kotler and Keller (2016) in their seminal marketing management textbook underscored that effective market segmentation must be measurable, accessible, and actionable. Their framework guides the interpretation of clusters made by machine learning methods. But their work was conceptual in nature

and did not provide a computational or algorithm approach to achieved these segmentation criteria, leaving a methodological gap that machine learning methods such as K-Means are well-placed to address.

3 Methods

This project followed a structured pipeline containing a data collection, preprocessing, clustering, and evaluation. Each action was designed to systematically transform raw customer data into meaningful segments that could influence business decisions.

3.1 Dataset

The dataset Mall Customer Segmentation from kaggle was used in this project. The dataset has a 200 records with five attributes: CustomerID, Gender, Age, Annual Income (thousands of dollars), and Spending Score (ranging from 10 to 100). The spending Score was a metrics assigned by the mall based on each purchasing of a customer's behavior and transaction history. The dataset Mall Customer Segmentation was selected because it provided a balanced mix of demographic and behavioral features that were suited for a unsupervised clustering analysis in retail context.

3.2 Data Preprocessing

Multiple preprocessing steps were performed to prepare the data for clustering. First steps is that the dataset was inspected for missing values but none were found, so no imputation was required. Second, the non-informative CustomerID column was removed, as it served only as a identifier and contributed no meaningful information to the process of clustering. Third, the categorical Gender features was formatted into numerical values using label encoding, where Female was formatted as 0 and Male as 1, enabling its inclusion in distance-based calculations, And last the numerical features: Age, Annual Income, and Spending Score, it is standardized using Min-Max Normalization, which rescaled all values to a range of 0 to 1. This guaranteed that no single features dominated the distance calculations in K-Means due to differences in scale.

3.3 Learning Algorithm

K-Means Clustering was implemented using the scikit-learn library in python. The algorithm sectioned the dataset into k cluster by minimizing the Within-Cluster Sum of Squares (WCSS) this is also known as inertia, which measured the compactness of each clusters. The optimal number of clusters was determined using two method:

- **Elbow Method -** WCSS was applied plotted against increasing values of k ,from 1 to 10. The point at which the rate of decrease sharply slowed, forming an "elbow", was identified as optimal k.
- **Silhouette Score -** For each candidate value of k, the silhouette score was computed to measure how similar each data point was to its own cluster compared to other clusters. A higher score for silhouette only indicated the clusters is defined better.

To improve the clustering stability and avoid poor minima, the K-Means++ initialization strategy was used, which selected a initial centroids in a way that spread them across the data space before the iterative optimization began. The algorithm was tested with multiple random seeds and the best results was obtained.

In this project two feature combination were examined to identify the most informative or important segmentation.

1 Annual Income versus Spending Score, and **2** Age versus Spending Score. These combination was chosen based on prior literature suggesting that income and spending behavior, as well as the age and spending patterns, were the most discriminative features for profiling customer in retail environment.

3.4 Evaluation Tools

The tools that will be use in cluster quality will be evaluated using Inertia that will measure the compactness of clusters, Silhouette Score that assess cluster cohesion and separation, and the Visual Inspection through 2D scatter plot using Matplotlib and Seaborn. The result of customer segments will be profiled and interpreted with actionable business insights. All implementation will be done in python using jupyter notebook, with libraries including pandas, numpy, scikit-learn, matplotlib, and seaborn.

4 Experiments/Results/Discussion

The main goal of this experiment was to carry out customer segmentation using the K-Means Clustering algorithm. We worked with the Mall Customer Dataset, aiming to pinpoint naturally occurring groups based on demographics and shopping habits.

4.1 Hyperparameters

To make sure the model was both stable and accurate, we made several important choices regarding hyperparameters and architecture:

Learning approach: we utilized the unsupervised K-Means algorithm from the sklearn.cluster library. This distance-based method organizes data into k clusters, with each point assigned to the cluster that has the closest mean.

Initialization strategy:(init='k-means++') This is important because standard K-Means can be quite sensitive to where the centroids are initially placed. By using k-means++, we choose initial centers that are strategically spaced apart, which helps speed up convergence and greatly lowers the chances of the algorithm getting trapped in a suboptimal solution.

Robustness: The algorithm was set up to run 10 times, each time using different random centroid seeds. The final model chosen is the one that produced the best results by minimizing inertia, which is measured by the Within-Cluster Sum of Squares.

Reproducibility: To make sure that the experiments can be checked and replicated with the same results, we used a fixed random state.

Feature Pre-processing (Min-Max Scaling): K-Means uses Euclidean distance as its foundation. Since "Annual Income" (which can go up to 137k) has a much wider numerical range compared to "Age" or "Spending Score," it would unfairly skew the distance calculations if we didn't scale it. To tackle this, we employed **Min-Max Normalization** with MinMaxScaler, which adjusts all features to a consistent range of [0, 1]. This way, every feature has an equal say in the clustering process.

4.2 Perfomance Metrics

We used two main quantitative metrics to assess how well our clustering performed and to figure out the best number of segments.

Inertia (Within-Cluster Sum of Squares - WCSS): This metric shows how closely the data points are grouped within their clusters. It's determined by adding up the squared distances from each data point to the center of its assigned cluster.

Silhouette Score: This evaluates the quality of the clusters by looking at how closely an object relates to its own cluster in comparison to other clusters.

- **Score near +1:** It shows that the sample is quite distant from the nearby clusters.
- **Score near 0:** This suggests that the sample is right on or extremely close to the decision boundary separating two adjacent clusters.

4.3 Quantitative Results and Optimal Cluster Selection

We explored various k values ranging from 2 to 10 to determine the best segmentation approach. You can find a summary of our findings in the table below:

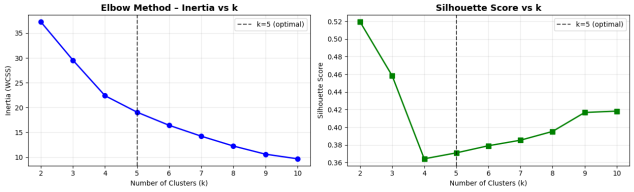


Figure 1: Figure 1: Elbow Method (WCSS) and Silhouette Score vs. k · confirms k=5 as optimal

To choose the final model, we used the Elbow Method. By plotting Inertia against the number of clusters, we pinpointed the "elbow point"—that moment when the decrease in inertia starts to slow down significantly. While k=2 gave us the highest Silhouette Score (0.519), it was too broad for effective marketing insights. So, we decided on k=5 as the best option. At k=5, the inertia is low enough (19.063), and the Silhouette Score shows a slight improvement (0.371), which suggests we have well-defined segments that can drive business actions.

4.4 Qualitative Results and Discussion

By looking at the average values of each of the five clusters, we were able to create unique customer personas. This qualitative analysis paves the way for more focused business strategies.

- **Cluster 1 (Diamond Customers):** When it comes to loyalty programs, the spotlight should definitely be on customers with a high annual income and a high spending score. These shoppers are the mall's biggest profit drivers, making them the key audience to target.
- **Cluster 4 (The "Sensei" / Cautious Wealthy)** It's interesting to see people with a high annual income but a low spending score. These folks have the money, yet they don't seem to engage much with luxury brands. This presents a unique opportunity for marketers in the high-end luxury sector to tap into this potential growth market.
- **Cluster 0 (Average Customers):** Middle-income earners who tend to spend moderately make up the biggest and most stable group of shoppers at the mall.
- **Cluster 3 (Young Moderate Spenders):** A younger crowd that might not have a lot of cash to throw around, but they sure know how to spend what they do have.
- **Cluster 2 (Low Income, Low Spenders):** Customers who might not have a lot of financial flexibility and aren't very engaged with the mall experience.

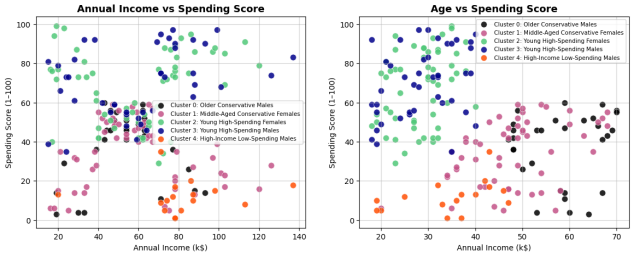


Figure 2: Figure 2: Cluster scatter plots · Annual Income vs. Spending Score (left) and Age vs. Spending Score (right), k=5.

4.5 Overfitting and Failure Analysis

One common pitfall of K-Means is overfitting, which happens when we set the number of clusters too high. This can lead the model to pick up on noise instead of the real patterns in the data. To

tackle this issue, we used the Silhouette Score to make sure that adding more clusters wouldn't create overlapping groups.

Another potential drawback of K-Means is its assumption that clusters are spherical and roughly the same size. If the data has elongated or irregular shapes, K-Means might misclassify points that are on the edges. However, our visualizations show that the segments in the Mall Customer Dataset are quite distinct, especially when we look at Annual Income versus Spending Score. Plus, by normalizing the data, we helped the model avoid being skewed by features with larger magnitudes.

5 Conclusion/Future Work

This report dives into how the **K-Means Clustering algorithm** was used for customer segmentation, specifically with the Mall Customer Dataset. The main goal was to step away from the old-school, rule-based methods that often falter when faced with the complexities of real-world data. Instead, we embraced a more systematic, data-driven approach. By examining factors like age, gender, annual income, and spending scores, we were able to pinpoint five unique customer segments. This journey included thorough data preprocessing, which involved getting rid of non-informative identifiers, applying label encoding for categorical variables, and using Min-Max Normalization to make sure that features like income didn't skew the distance-based calculations.

The K-Means algorithm, especially when paired with the **K-Means++ initialization** method, turned out to be the most effective choice for this task. We went with this setup because it helps avoid the algorithm getting trapped in less-than-ideal solutions by carefully spacing out the starting centroids. To confirm its performance, we used the **Elbow Method** and **Silhouette Scores**. Interestingly, while k=2 gave us the highest Silhouette Score, we opted for k=5 as the best number of clusters. This choice allowed for a more detailed and useful breakdown of our customer base, helping us pinpoint specific personas like "Diamond Customers" (those with high income and spending) and "Cautious Wealthy" individuals.

In this situation K-Means is the highlight, after normalizing the dataset's features, we ended up with fairly distinct and spherical groups that fit well with what the algorithm expects. Feature scaling played a crucial role here; it acted as an essential adjustment that kept the wide range of annual income from overshadowing other demographic details like age during the clustering process. Plus, by running 10 iterations with different random seeds, we made sure that our final model was both strong and consistent.

Looking ahead, explore using Principal Component Analysis (PCA). This could help us see if reducing dimensions might improve how well we can separate clusters, especially if we expand our dataset to include more intricate variables. We also think it would be beneficial to conduct a comparative study with Hierarchical Clustering. By assessing both methods using quantitative metrics like the Silhouette Score, we could find out if a dendrogram-based approach provides more stable or interpretable segments, particularly in retail settings where customer behavior isn't as clear-cut.

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